SETTribe
An Ed-Tech Prudentia

SETTribe LLP

Employee Leave Management System

Deadline: 17 August 2026 (within 1 week) **Duration:** 5–6 hours

1. Assessment Objective

Build a web application where employees can register/login, apply for leave, view their leave history and manage leave requests through a simple workflow.

2. Technology Requirements

Frontend	React.js, React Router, Axios/Fetch, CSS or Bootstrap
Backend	Java 17+, Spring Boot, Spring Web, Spring Data JPA
Database	MySQL
Authentication	JWT or session-based authentication
Version Control	Git + GitHub

3. Functional Requirements

- User registration, login and logout.
- Protected dashboard.
- Apply for leave with type, dates and reason.
- View personal leave requests and status.
- Edit/cancel a pending request.
- Admin can approve/reject requests.
- Filter requests by status or leave type.

4. Database Requirements

users: id, name, email, password, role, created_at

leave_requests: id, user_id, leave_type, from_date, to_date, reason, status, created_at

Use suitable primary keys, foreign keys, constraints and data types.

5. Backend API Requirements

Method	Endpoint	Purpose
POST	/api/auth/register	Register
POST	/api/auth/login	Login
GET	/api/leaves	List leaves
POST	/api/leaves	Apply leave
PUT	/api/leaves/{id}	Update request
DELETE	/api/leaves/{id}	Delete request
GET	/api/admin/leaves	Admin list
PUT	/api/admin/leaves/{id}/status	Approve/reject

Use Controller → Service → Repository structure. Validate requests, return suitable HTTP status codes, protect APIs and prevent unauthorized access.

6. React Frontend Requirements

- Login and registration pages.
- Protected dashboard.
- Forms for CRUD operations.

- API integration with loading, success, empty and error states.
- Basic responsive UI.

7. Authentication & Validation

- Registration and login must work.
- Invalid credentials must show an error.
- Passwords must not be stored as plain text.
- Protected APIs must reject unauthenticated requests.
- Required fields must be validated.
- Invalid IDs/request data must be handled.
- Users must not access another user's data.

8. SQL Assessment

Write and execute the following basic SQL queries:

- Display all users.
- Display leave requests for a user.
- Count requests by status.
- Find pending requests.
- Find approved requests ordered by date.
- Join users with leave requests.
- Update request status.
- Delete a pending request.

9. Git & GitHub Requirements

- Create a GitHub repository.
- Use meaningful commit messages.
- Keep frontend and backend organized.
- Add README.md with setup/run instructions.
- Do not upload passwords or secrets.

10. Assessment Marks — 100

Category	Marks
Java + Spring Boot	20
React	20
Authentication & Authorization	15
REST API + CRUD	15
MySQL + SQL	10
Validation + Error Handling	5
Git + GitHub	5
Code Quality + UI	5
Project Explanation	5
Total	100

11. Practical Assessment Tasks

Task 1 — Project Setup + Database

Set up the React frontend, Spring Boot backend and MySQL database. Ensure the project runs successfully.

Task 2 — Registration + Login

Implement registration, login, authentication and logout with validation and password security.

Task 3 — Core Business Features

Implement the main CRUD/business functionality described for **Employee Leave Management System**.

Task 4 — Project-Specific Workflow

Implement the project-specific workflow, filters, summary or role-based functionality.

Task 5 — React + API Integration

Create the required React screens and connect them with the Java REST APIs.

Task 6 — SQL + GitHub + README

Complete the SQL queries, push the project to GitHub and add a README with setup instructions.

12. Final Submission

- GitHub repository URL.
- Working React application.
- Working Spring Boot application.
- MySQL database/schema.
- SQL query file.
- README.md with setup instructions.
- Working demonstration.

13. Completion Checklist

- ■ Registration and Login working
- ■ Authentication/protected routes working
- ■ Main features working
- ■ MySQL connected
- ■ SQL queries completed
- ■ React connected with APIs
- ■ GitHub repository submitted
- ■ README completed
- ■ Project demonstrated